

Sujay Seeram

Cyber-Physical Systems, Embedded Electronics & Physics-Informed Engineering

Email: sujayat2007@gmail.com | GitHub: github.com/seeramsujay | Portfolio: seeramsujay.github.io | Location: Visakhapatnam / Coimbatore, India

ENGINEERING PROFILE & CORE PHILOSOPHY

Electronics and Communication Engineering undergraduate at Amrita Vishwa Vidyapeetham specializing in **resource-constrained cyber-physical systems**, **embedded hardware architecture**, and **physics-informed computing**. An open, execution-oriented learner and quick adapter driven by the “**Extreme Economic Engineering**” paradigm: deriving minimalist hardware configurations from physical first principles to drastically slash physical component counts, quiescent power consumption, and bill-of-materials (BOM) while maximizing system fidelity. Beyond core engineering, deeply committed to applying technological innovation toward **renewable energy**, **environmental sustainability**, and **social responsibility**. Disciplined Naval NCC cadet and 3-time Services Selection Board (SSB) attendee.

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Core Coursework:** Microprocessors & MCUs (ARM Cortex/8051), Signals & Systems, Control Engineering, Digital System Design, Analog Electronic Circuits, VLSI Design, Electromagnetic Theory, Communication Networks, Power Electronics.
 - **Academic Focus:** Cyber-Physical Architectures, Acoustic/Vibration Signal Processing, Embedded Edge ML, Sensor Conditioning.
- **Navy Children School (NCS), CBSE** Visakhapatnam, India
Senior Secondary (Class XII): 92.5% | Secondary School (Class X): 95.0% *2021 – 2023*

TECHNICAL SKILLS

- **Silicon, SoCs & MCUs:** STM32 (ARM Cortex-M4), ESP32 (Dual-Core Xtensa, RISC-V ULP), Raspberry Pi, RP2040, ATmega328P.
- **Firmware & Low-Level Software:** Embedded C, C++17/20, FreeRTOS, TinyML, Python, Assembly (ARM/x86), Linux Device Drivers.
- **Hardware Protocols & Buses:** SPI, I2C, UART/USART, CAN bus, PWM, ADC/DAC, DMA-driven waveform synthesis, GPIO edge interrupts.
- **Analog Conditioning & Sensors:** OPA1612 Butterworth active filters, ACS712 current, ZMPT101B voltage, MPU6050 6-DoF IMUs.
- **DSP, Signal Processing & Edge AI:** Fast Fourier Transform (FFT) spectral analysis, MFCC extraction, Kalman filtering, Isolation Forest, TFLite Micro.
- **Prototyping & Instrumentation:** KiCad PCB layout & schematic design, Digital Storage Oscilloscopes (DSOs), Logic Analyzers, JTAG/SWD.

ACHIEVEMENTS, DISTINCTIONS & MILITARY DISCIPLINE

- **Rapid Prototyping Portfolio & Engineering Track Record** *Portfolio Metrics*
 - **95 Projects Explored, 50 Executed, 3 Major Hackathons Won:** Interdisciplinary distribution of **15 AI+CS+ECE** cyber-physical projects, **25** dedicated ECE/embedded systems, **5** core CS/systems software, and **24** foundational sandbox builds.
 - **Real-Time Production Deployment (SAFE):** Engineered **SAFE**, an operational, production-grade warehouse asset tracking ecosystem featuring low-power BLE duty cycling and high-speed batch scanning (*details reserved for interview discussion*).
- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at the prestigious IIT Madras Smart India Challenge for engineering economical cyber-physical solutions.
 - Honored as the **Youngest Panelist** at the summit, presenting on hardware miniaturization and extreme economic component engineering.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - **NCC Senior Division:** Trained cadet in **3(A) Naval Unit NCC, Visakhapatnam (VSP)**; developed crisis leadership, nautical discipline, and obstacle course proficiency.
 - **Services Selection Board (SSB):** Attended 3 SSB selection boards (**Screened-in twice & Conferenced-out twice**; Screened-out once); demonstrated resilient situational decision-making and psychological adaptability under high-stress evaluations.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at the **GARC Chennai Meet**; engaged with automotive engineers on sensor failure preemption, non-invasive telemetry, and embedded homologation standards.

FEATURED ENGINEERING PROJECTS (ORDERED PORTFOLIO)

I. Interdisciplinary Cyber-Physical AI (AI, CS & EC Combined)

- **1. EVolvAI – Generative AI & Physics-Constrained EV Grid Optimization** *Release v1.0.0*
 - Architected a scalable cyber-physical demand framework in PyTorch coupling a **Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm to simulate charging loads under distribution grid bus limits and climate stress anomalies.
 - Achieved $R^2 \approx 0.5$ predictive accuracy on sparse field data; built an interactive Streamlit simulation console for municipal engineers.
- **2. MotorSafe – Tri-Modal Edge-AI Predictive Maintenance Ecosystem** *Release v0.01*
 - Constructed an edge-native telemetry pipeline fusing 3 synchronized sensor channels: **ACS712** current, **ZMPT101B** voltage, and **MPU6050** vibration.
 - Engineered on-chip FFT DSP to extract spectral sidebands and deployed an unsupervised **Isolation Forest** on ESP32 to preempt bearing breakdown without cloud latency, streaming live diagnostics to an interactive Three.js 3D digital twin dashboard.
- **3. SpecRAG – Zero-Hallucination Firmware & Datasheet Auditing Gateway** *Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon datasheets with embedded C/C++ firmware using the **Model Context Protocol (MCP)**.
 - Performed localized vector retrieval against indexed silicon register banks to preemptively catch peripheral pin collisions, clock-tree prescaler errors, and timing hazards directly in VS Code before compilation.

II. Embedded Systems & Electronics (ECE)

- **4. AQUAPULSE – Cognitive Software-Defined Sonar Payload for AUVs** *Repository*
 - Engineered a cognitive software-defined sonar payload for Autonomous Underwater Vehicles (AUVs) meeting the SIH26058 MoES mandate.
 - Implemented zero-CPU DMA-to-DAC acoustic synthesis on **STM32 / ESP32-S3** with **OPA1612** active filtering, deploying an INT8 **TinyML** engine to dynamically adapt ping frequency (100–480 kHz) across shifting thermoclines, slashing acoustic power consumption by **38%**.
- **5. Pheno-Speak – Edge Acoustic Vocal Biomarker Extraction Engine** *Repository*
 - Built an edge audio digital signal processing pipeline for non-invasive vocal biomarker extraction aligned with SDG 3.4 (Digital Phenotyping).
 - Extracted MFCCs, jitter, and shimmer features on lightweight microcontroller hardware without offloading unencrypted audio to external endpoints.
- **6. Dust-Vision & Dust-Sim – Edge Particulate Tracking & Dispersion Engine** *Release v0.1*
 - Deployed computer vision models on edge camera modules to quantify real-time construction site particulate pollution and dust plumes.
 - Coupled sensor data with a custom particle kinematics physics simulation (**dust-sim v1.0.0**) modeling dispersion under thermal and wind shifts.

III. Systems Software, Developer Tools & Computer Science (CS)

- **7. ytstr – High-Performance Psychoacoustic Audio Engine** *Release v2.0.0*
 - Developed a terminal-native streaming runtime powered by a psychoacoustic **DecisionEngine** executing real-time spectral analysis.
 - Implemented 8 broadcast-grade transition algorithms and built an active backpressure memory pool, slashing runtime RAM by **70%** (< 260MB).
- **8. Informed-Poll – Swipeable Civic Intelligence Platform** *Release v1.0.0*
 - Developed a mobile-first civic intelligence companion featured in the Google Solution Challenge, pairing swipe mechanics with verified candidate records.
 - Integrated localized neural RAG using **LanceDB** and Gemini 3.1 Flash Lite over a Cloud Run FastAPI backend with zero-hallucination verification.
- **9. Kavach-UPI – Android Accessibility Security Shield** *Repository*
 - Authored an on-device security shield leveraging Android Accessibility APIs to detect active screen-sharing malware during financial UPI transactions.
 - Instantly deployed an opaque system window overlay to block remote touch injections while triggering localized security alerts.